

Here are 5 MCQs based on the context:

Q1: What is the definition of half-life?

- A) The time taken by a radioactive material to disintegrate completely
- B) The time taken by a radioactive material to disintegrate to half of its original value
- C) The rate at which a radioactive element decays
- D) The term used to describe the decay of radioactive materials

Answer: B

Explanation: The context defines half-life as the time taken by a radioactive material to disintegrate to half of its original value.

Q2: What fraction of a radioactive sample has decayed after two half-lives have elapsed?

- A) $\frac{1}{4}$
- B) $\frac{1}{2}$
- C) $\frac{3}{4}$
- D) $\frac{2}{3}$

Answer: C

Explanation: According to the context, after two half-lives, $\frac{3}{4}$ of a radioactive sample has been decayed.

Q3: What is the half-life of carbon-14?

- A) 1000 years
- B) 5730 years
- C) 10000 years

D) 1 million years

Answer: B

Explanation: The context states that carbon-14 has a half-life of 5730 years.

Q4: If a radioactive sample has a half-life of 100 years, how much of the sample will remain after 200 years?

A) $\frac{1}{4}$

B) $\frac{1}{2}$

C) $\frac{1}{3}$

D) $\frac{3}{4}$

Answer: A

Explanation: After one half-life (100 years), half of the sample remains. After another half-life (200 years total), half of the remaining half will decay, leaving $\frac{1}{4}$ of the original sample.

Q5: How do the half-lives of various radioisotopes compare?

A) They are all equal

B) They range from a few microseconds to billions of years

C) They are all measured in years

D) They are all measured in microseconds

Answer: B

Explanation: The context states that half-lives for various radioisotopes can range from a few microseconds to billions of years.

These questions cover 2 recall (Q1, Q3), 2 application (Q4, Q2), and 1 analysis (Q5) questions according to Bloom's taxonomy. The answers and explanations are verifiable from the context.